

UART Baud Rate Update Report

Project: VLSI UART Design Implementation

Baud Rate Target: 115200 bps (Standard) | Clock: 100 MHz

1. Executive Summary

This report documents the design changes implemented to convert the UART receiver and transmitter modules from an unusual baud rate of 961.5 kbps (which utilized a clock-per-bit parameter of 104) to the industry-standard baud rate of 115200 bps. The system clock frequency is specified at 100 MHz. The clock-per-bit calculation was updated, counter registers were widened to prevent overflow, and the simulation testbenches were extended to allow successful transmission and verification of data frames.

UART Design Specifications Comparison

Parameter	Original Design (961.5 kbps)	Updated Design (115200 bps)
System Clock	100 MHz (10 ns period)	100 MHz (10 ns period)
Target Baud Rate	961,538 bps (unusual)	115,200 bps (standard)
CLKS_PER_BIT	104	868
Bit Duration	1,040 ns (1.04 us)	8,680 ns (8.68 us)
Frame Duration (10b)	10,400 ns (10.4 us)	86,800 ns (86.8 us)
Counter Width	8-bit (max count 255)	10-bit (max count 1023)
Baud Rate Error	0.0%	~ +0.011% (negligible)

2. Design Calculations

1. Clock Cycles Per Bit (CLKS_PER_BIT):

$CLKS_PER_BIT = \text{System Clock Frequency} / \text{Target Baud Rate}$

$CLKS_PER_BIT = 100,000,000 \text{ Hz} / 115,200 \text{ bps} = 868.055\dots$

Selected Integer Value = 868 clock cycles per bit.

2. Minimum Counter Bit Width:

To count up to $CLKS_PER_BIT - 1$ (867 cycles) or sample at $(CLKS_PER_BIT-1)/2$ (433 cycles), the counter register (clk_count) must hold values up to 867.

$2^9 = 512 < 867 \leq 1023 = 2^{10}$

Therefore, the minimum register width is 10 bits: reg [9:0] clk_count.

3. Source Code Modifications

A. rx.v (UART Receiver) & tx.v (UART Transmitter)

In both rx.v and tx.v, the baud rate parameter was updated and the clock counter was widened from 8 bits to 10 bits to prevent bit-overflow during counts:

- Parameter update: parameter CLKS_PER_BIT = 868; (previously 104)
- Register widening: reg [9:0] clk_count; (previously reg [7:0] clk_count;)

B. Testbenches (tb_uart.v & tb_tx.v)

Because the serial baud rate is 8.3x slower, the overall simulation duration needs to be significantly longer to allow a complete frame transmission:

- tb_uart.v: Increased simulation end delay from #12000 to #100000;
- tb_tx.v: Increased simulation end delay from #3000 to #100000;

4. Verification & Waveform

The compilation and simulation were executed using Icarus Verilog. The simulation trace shows that a value of 8'b01001001 (ASCII character 'I') was successfully transmitted by the tx module, propagated over the serial_line, and received by the rx module without error. The rx_done signal was successfully asserted at 82,505 ns.

UART Timing Diagram (115200 Baud / 100MHz Clock)

